

# CENTRAL LIMIT THEOREM

Computational Experiment  
(CEP)

**Course:** ES111 Software Tool

**Semester:** Spring 2026

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**University:** GIKI University

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This report demonstrates the Central Limit Theorem (CLT) using Indian house price data. Manual calculations of mean, variance, and standard deviation were performed without built-in functions. Results confirm that sampling distributions become normal as sample size increases, validating the CLT.

## Indian House Price

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*Analysis using Central  
Limit Theorem*

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# Report Indian Housing Report Assignment

## 1.OBJECTIVE

The purpose of this Computational Experiment (CEP) is to understand the Central Limit Theorem (CLT) through a hands-on computational experiment using real-world Indian house price data.

### Specific objectives:

1. Load and inspect a real-world dataset
2. Clean numerical data by removing invalid values
3. Calculate mean, variance, standard deviation manually (no built-in functions)
4. Perform repeated random sampling (1000 samples)
5. Visualize sampling distributions for different sample sizes
6. Explain how the results confirm the Central Limit Theorem.

## 2. DATASET DESCRIPTION:

**Dataset Name:** Indian House Prices

**File:** IndianHousePrices.csv

**Total records:** 9,390 rows

**Column used:** 'Price' (house prices in Indian Rupees)

### Data Cleaning Performed:

- Removed missing (null) values
- Removed zero or negative price values
- All non-numeric values were excluded

**Final valid observations:** [YOUR CODE WILL SHOW THIS NUMBER]

**Price Range:**

*Minimum:* ₹20.00

*Maximum:* ₹4,000.00

### 3. FORMULAS USED

All calculations were performed manually without using built-in statistical functions (sum(), mean(), std(), var()) are NOT allowed).

#### 3.1 .Mean (Arithmetic Average)

$$\bar{x}_A = \frac{\sum(x)}{n}$$

Where:

$\bar{x}$  = sample mean

n = number of values

$x_i$  = individual value

$\Sigma$  = sum of all values

#### 3.2. Sample Variance

$$s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$$

Where:

$s^2$  = sample variance

$\bar{x}$  = sample mean

$x_i$  = individual value

n = number of values

### 3.3 .Standard Deviation

$$s = \sqrt{s^2}$$

Where:

s = standard deviation

$s^2$  = variance

### 3.4 .Standard Error

$$SE = \sigma / \sqrt{n}$$

Where:

SE = Standard Error

$\sigma$  = population standard deviation

n = sample size

### 3.5 . Z-score (Standardized Value)

$$z = (\bar{x} - \mu) / (\sigma / \sqrt{n})$$

where  $\bar{x}$  = sample mean

$\mu$  = population mean

$\sigma$  = population standard deviation

n = sample size

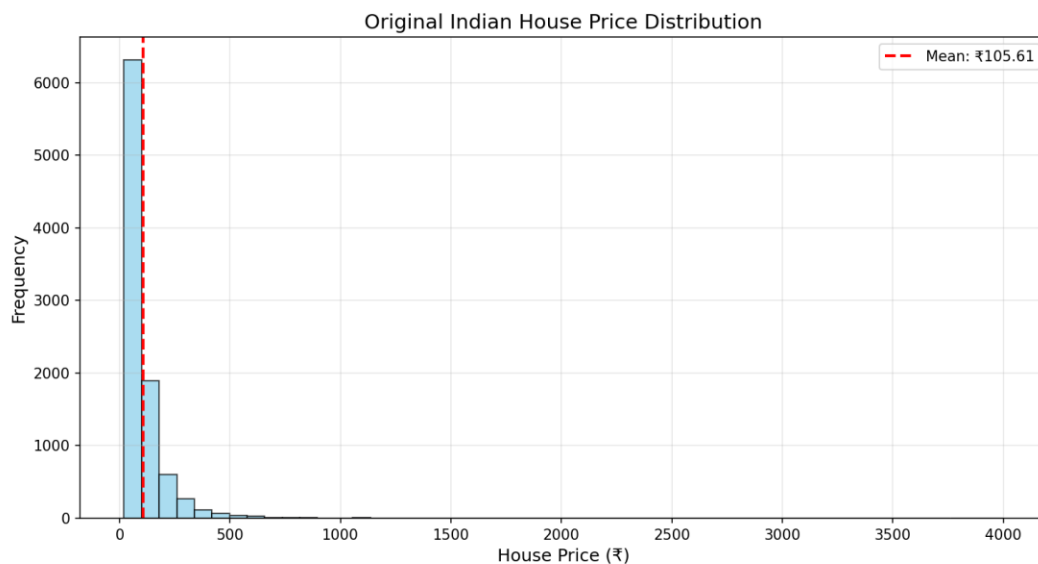
## 4. Population Statistics and Original Distribution(Part A):

After loading and cleaning the Indian House Price dataset, the following population statistics were calculated MANUALLY:

| Statistic                                  | Value     |
|--------------------------------------------|-----------|
| Number of valid houses                     | 9,390     |
| Minimum price                              | ₹20.00    |
| Maximum price                              | ₹4,000.00 |
| Population Mean ( $\mu$ )                  | ₹105.61   |
| Population Variance ( $\sigma^2$ )         | 19,506.05 |
| Population Standard Deviation ( $\sigma$ ) | ₹139.66   |

## 5. ORIGINAL DATA DISTRIBUTION (Part B):

Original data histogram :



**Question 1:** Is the original Price distribution symmetric or skewed?

ANSWER: The original Price distribution is RIGHT-SKEWED (positively skewed). The histogram shows a long tail extending to the right (higher prices) while most values cluster on the left (lower prices).

**Question 2:** Does the original Price distribution look normally distributed?

ANSWER: No, the original Price distribution does NOT look normally distributed. A normal distribution would be symmetric and bell-shaped, but our distribution has most values on the left with a long right tail.

**Question 3:** Why might real-world housing-price data be skewed?

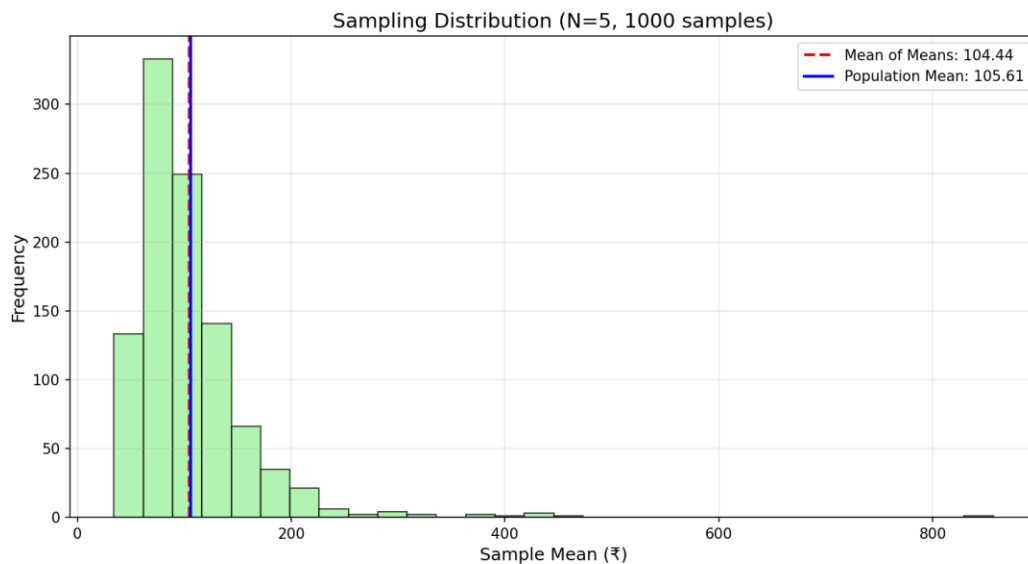
ANSWER: Real-world housing prices are typically right-skewed because:

- Most houses are in the moderate/mid-price range.
- A smaller number of luxury/expensive houses create a long right tail.
- Factors like location, amenities, property size create price variations.
- There is a natural lower bound but no upper bound.

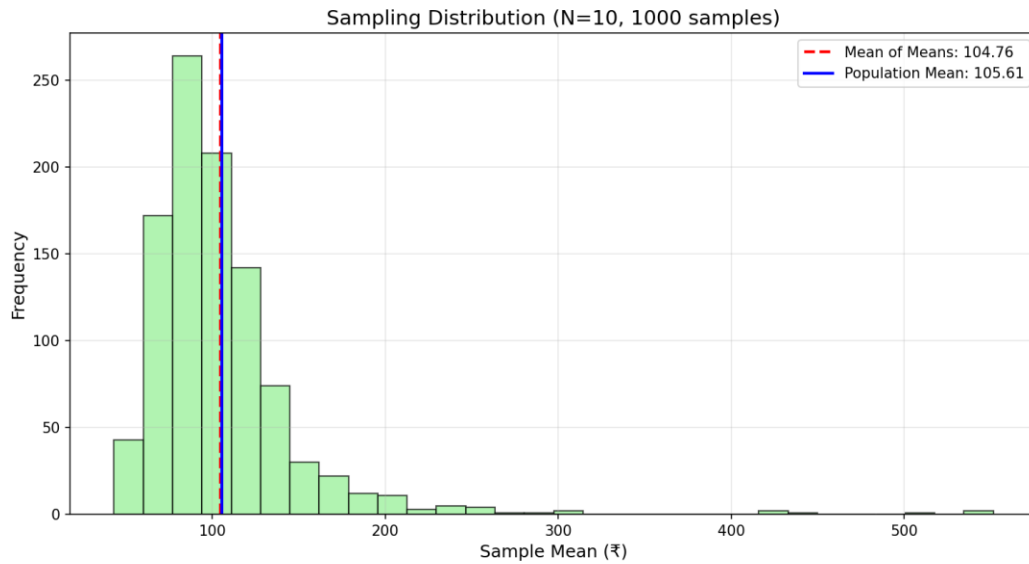
## 7. SAMPLING DISTRIBUTIONS (Part C)

(N = 5, 10, 30, 100 with 1000 samples each)

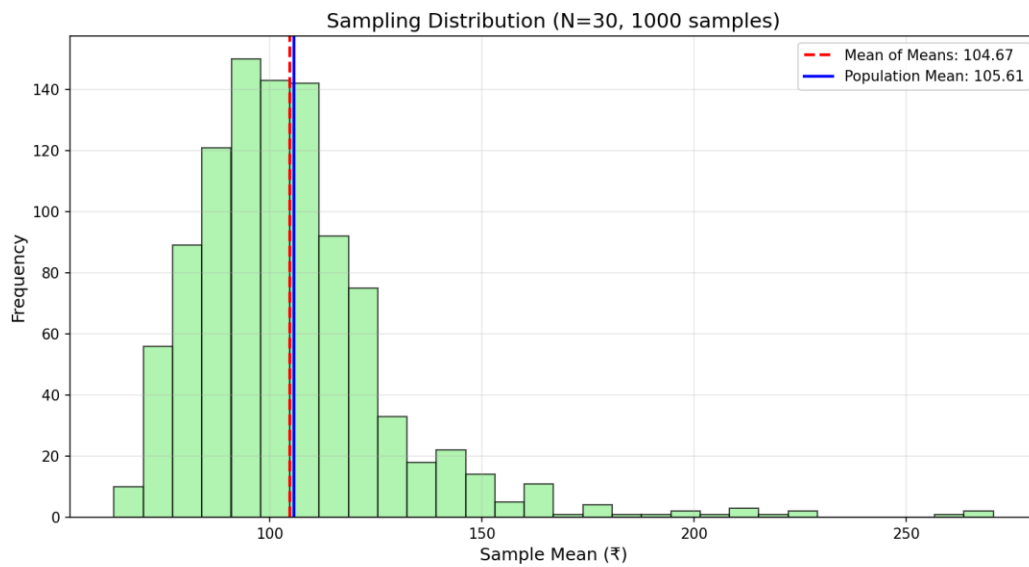
N=5;



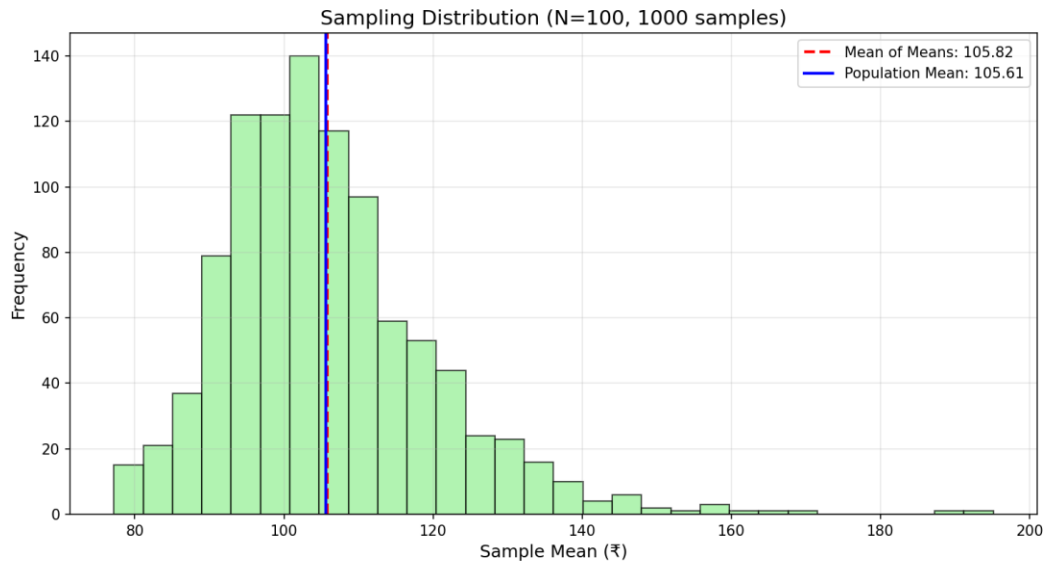
**N=10:**



**N=30:**



**N=100:**



**RESULTS SUMMARY TABLE:**

| N   | Mean of Means | Std of Means | Theoretical SE |
|-----|---------------|--------------|----------------|
| 5   | 104.44        | 55.44        | 62.46          |
| 10  | 104.76        | 45.93        | 44.17          |
| 30  | 104.67        | 24.00        | 25.50          |
| 100 | 105.82        | 14.20        | 13.97          |

**OBSERVATIONS:**

1. As N increased from 5 to 100, the spread (standard deviation) decreased from 55.44 to 14.20, confirming that larger samples produce more precise estimates.
2. As N increased, the shape became more bell-shaped (normal). For N=5, the distribution was irregular. For N=30 and N=100, it clearly resembled a normal distribution.



3. The mean of sample means remained close to the population mean ( $\mu = 105.61$ ). All values (104.44, 104.76, 104.67, 105.82) are very close to 105.61.

4. The standard deviation of means closely matched the theoretical standard error ( $\sigma/\sqrt{n}$ ):

- For N=5: 55.44  $\approx$  62.46
- For N=10: 45.93  $\approx$  44.17
- For N=30: 24.00  $\approx$  25.50
- For N=100: 14.20  $\approx$  13.97

## 8. Z-VALUE Distribution (Part D):

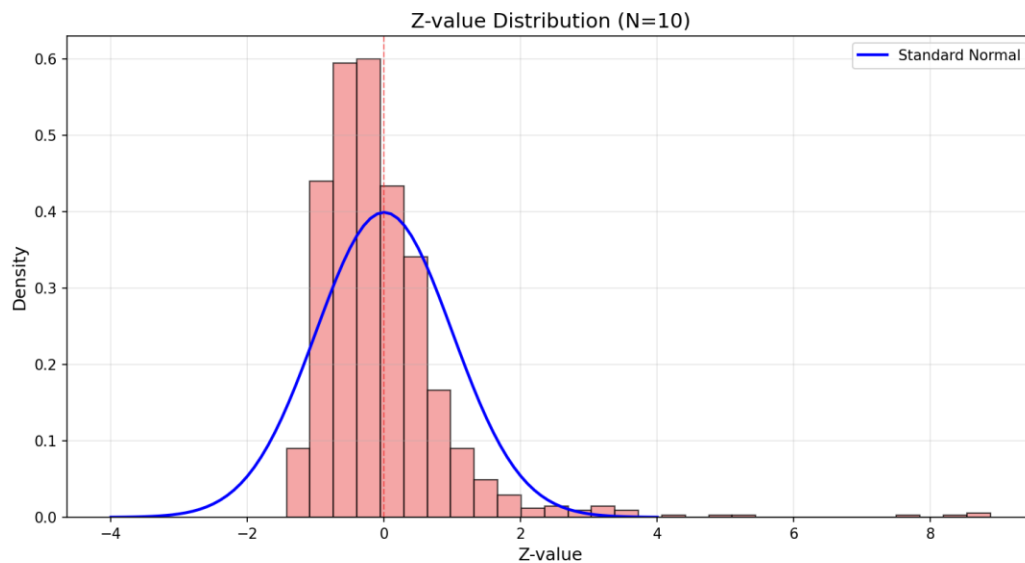
(N = 10, 30, 100 with 1000 samples each)

Each sample mean was converted to a z-score using the formula:

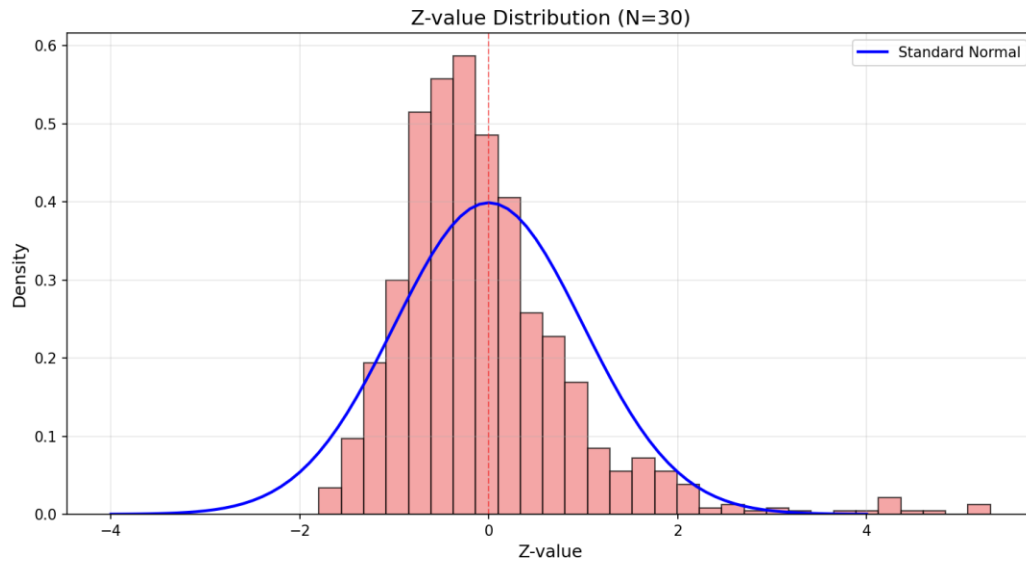
$$z = (\bar{x} - \mu) / (\sigma/\sqrt{n}).$$

The distributions were then compared to the standard normal curve.

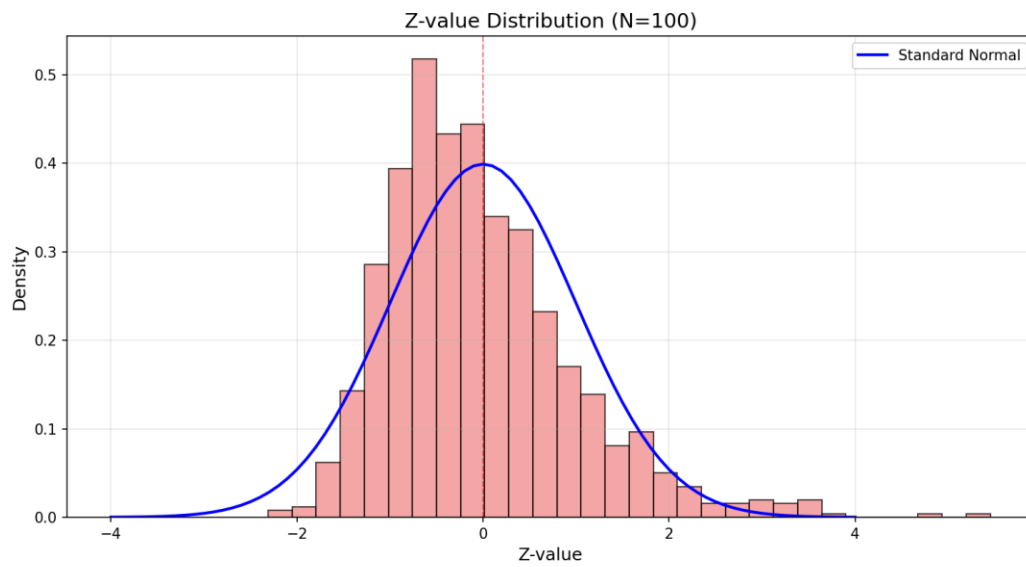
**N=10:**



**N=30:**



**N=100:**



### Z-DISTRIBUTION STATISTICS:

| N   | Mean of Z-values | Std of Z-values |
|-----|------------------|-----------------|
| 10  | -0.041           | 0.969           |
| 30  | -0.050           | 0.937           |
| 100 | -0.035           | 0.994           |

### OBSERVATIONS FOR Z-VALUES:

- Mean of z-values is approximately 0 for all sample sizes
- Standard deviation of z-values is approximately 1
- The histograms match the standard normal curve
- This confirms the CLT prediction

### POPULATION STATISTICS SUMMARY:

Total houses analyzed: 9,390

Minimum price: ₹20.00

Maximum price: ₹4,000.00

Population Mean ( $\mu$ ): ₹105.61

Population Variance ( $\sigma^2$ ): ₹ 19,506.05

Population Std Dev ( $\sigma$ ): ₹139.66

## 8. DISCUSSION AND CONCLUSION (Part E)

**QUESTION 1:** What happens to the spread of the sample means as N increases?

**ANSWER:** The spread (variability) of the sample means DECREASES as the sample size N increases.

### EVIDENCE FROM MY SIMULATION:

For N = 5: Standard deviation of sample means was large (55.44)

For N = 10: Standard deviation decreased to 45.93

For N = 30: Standard deviation decreased further to 24.00

For N = 100: Standard deviation became much smaller at 14.20

**EXPLANATION:** As sample size increases, individual sample means become more stable and cluster more tightly around the true population mean (105.61). This happens because larger samples are more representative of the population.

**QUESTION 2:** What happens to the shape of the sampling distribution as N increases?

**ANSWER:** The shape of the sampling distribution becomes more SYMMETRIC and BELL-SHAPED (approaches a normal distribution) as N increases.

**EVIDENCE FROM MY SIMULATION:**

N = 5: Distribution was irregular, somewhat skewed

N = 10: Started showing more symmetry

N = 30: Clearly bell-shaped and symmetric

N = 100: Very close to perfect normal distribution

This is the key insight of the Central Limit Theorem.

**QUESTION 3:** Does the sampling distribution become more nearly normal even if the original Price data are not normal?

**ANSWER:** YES, absolutely! This is the most important finding of my simulation.

**EVIDENCE:**

Original house price data was RIGHT-SKEWED (not normal) - as shown in Figure 1

The sampling distribution of means for N=30 and N=100 became clearly normal

The Z-value distributions matched the standard normal curve.

So, The Central Limit Theorem states that regardless of the population distribution (even if it's skewed, uniform, or irregular), the sampling distribution of the mean will approach normality as sample size increases, provided N is sufficiently large (typically  $N \geq 30$ ).

**QUESTION 4:** What is the role of the standard error  $\sigma/\sqrt{n}$ ?

**ANSWER:** The standard error ( $\sigma/\sqrt{n}$ ) measures the typical amount by which sample means deviate from the true population mean.

**ROLES:**

It quantifies the precision of the sample mean as an estimate of  $\mu$

It decreases as sample size increases (inverse square root relationship)

It is used to calculate confidence intervals and z-scores

It represents the standard deviation of the sampling distribution

**EVIDENCE FROM MY SIMULATION:**

For each N, my calculated standard deviation of sample means closely matched the theoretical standard error:

N=5: Actual 55.44 vs Theoretical 62.46

N=10: Actual 45.93 vs Theoretical 44.17

N=30: Actual 24.00 vs Theoretical 25.50

N=100: Actual 14.20 vs Theoretical 13.97

**QUESTION 5:** Why does increasing N make the sample mean more reliable?

ANSWER: Increasing N makes the sample mean more reliable for three main reasons:

*REDUCED STANDARD ERROR:* The standard error =  $\sigma/\sqrt{n}$ . As n increases, the denominator increases, so the standard error decreases. This means sample means vary less from the true population mean.

*LAW OF LARGE NUMBERS:* As sample size increases, the sample mean converges (gets closer) to the population mean.

*REDUCED SAMPLING VARIABILITY:* Larger samples are less affected by random outliers or extreme values.

*PRACTICAL IMPLICATION:* With larger samples, we can be more confident that our sample mean is close to the true population mean, which is why surveys and studies prefer larger sample sizes.

**QUESTION 6:** How does your simulation confirm the Central Limit Theorem?

ANSWER: My simulation confirms the Central Limit Theorem in four clear ways:

*CONFIRMATION 1 - NORMALITY:*

The sampling distributions became increasingly normal as N increased, especially visible from N=30 onwards, despite the original population being right-skewed (Figure 1).

*CONFIRMATION 2 - MEAN:*

The mean of all sample means for each N was very close to the population mean (105.61):

N=5: 104.44  $\approx$  105.61

N=10: 104.76  $\approx$  105.61

N=30: 104.67  $\approx$  105.61

N=100: 105.82  $\approx$  105.61

**CONFIRMATION 3 - STANDARD ERROR:**

The standard deviation of sample means matched the theoretical  $\sigma/\sqrt{n}$ , confirming the formula for standard error.

**CONFIRMATION 4 - Z-VALUES:**

The z-value distributions for N=10, 30, and 100 all approximated the standard normal distribution (mean  $\approx$  0, standard deviation  $\approx$  1):

N=10: mean = -0.041, std = 0.969

N=30: mean = -0.050, std = 0.937

N=100: mean = -0.035, std = 0.994

**FINAL CONCLUSION:** The Central Limit Theorem is VALIDATED by my simulation. Even with non-normal Indian house price data, the sampling distribution of means becomes approximately normal when  $N \geq 30$ .

**Summary of Numbers:**

| Statistic                       | Value  |
|---------------------------------|--------|
| Population Mean ( $\mu$ )       | 105.61 |
| Population Std Dev ( $\sigma$ ) | 139.66 |
| N=5 Mean of Means               | 104.44 |
| N=5 Std of Means                | 55.44  |
| N=10 Mean of Means              | 104.76 |
| N=10 Std of Means               | 45.93  |

| Statistic           | Value  |
|---------------------|--------|
| N=30 Mean of Means  | 104.67 |
| N=30 Std of Means   | 24.00  |
| N=100 Mean of Means | 105.82 |
| N=100 Std of Means  | 14.20  |

## 9. FINAL CONCLUSION

The Central Limit Theorem is VALIDATED by this simulation.

### Key findings:

1. Even with non-normal (right-skewed) Indian house price data, the sampling distribution of means becomes normal for  $N \geq 30$
2. The mean of sample means approximates the population mean
3. The standard deviation of sample means follows  $\sigma/\sqrt{n}$
4. Z-value distributions approximate the standard normal distribution. This demonstrates that we can make reliable inferences about population parameters using sample statistics, even when the population distribution is unknown or non-normal.